Dear Editors,

We submit a Problem Solving Protocol: a reusable procedure for auditing cross-cohort feature-selection benchmarks on public transcriptome cohorts, applied in two further tissues against predictions that were written and hash-stamped before those data were read. It is not a biomarker discovery study and reports no new biomarker. The title is "Auditing cross-cohort feature-selection benchmarks: cohort composition and tissue-context-dependent confounding across kidney, liver and colon transcriptomes."

Methods for deriving reproducible signatures are usually compared on one fixed collection of public cohorts. The protocol asks whether such a comparison measures the methods or the cohorts, through four checks: a patient-overlap and procurement audit, a cohort-composition sensitivity analysis, a control-definition sensitivity analysis, and a correction benchmark. It condenses them into a nine-item reporting checklist (Table 4). The analyses in the manuscript are the application that shows what each check detects.

We think three results make it useful to your readership beyond the disease in which it was developed:

1. A benchmark verdict can be a property of the cohorts. On five diabetic kidney disease discovery cohorts, the reproducibility difference between a stability selector and a co-expression hub criterion ranged from -0.106 to +0.170 across three-cohort subsets, while the mean external AUROC advantage stayed positive.

2. A shared pre-analytical confounder becomes the most reproducible signal. Every discovery contrast compares needle biopsies with surgical tissue, and an immediate-early module tracked that contrast in 8 of 9 kidney datasets. Substituting a biopsy control retained only 14-56% of the naive signal across seven diagnoses; batch correction by cohort could not touch it.

3. Both findings were tested outside kidney against predictions fixed in advance. Before any liver or colon expression value was read, we hashed three predictions. Verdict instability recurred in liver but not in colon, whose cohorts agree closely about effects, and the confounder was most prominent among the most reproducible genes in kidney. Two predictions failed: in colon, where both arms are biopsies, the module rose with inflammation, so the confounding it indexes depends on tissue context. We report the failures in full, and accordingly present a control obtained by the same procedure as the design alternative to prefer where clinically defensible and available, not as a universal fix.

The audited pipeline declining to promote its own uncorrected candidates is the intended outcome of the application, not a shortfall. All data are public; the harmonised cohorts, audit database and code regenerating every figure and table are released at https://github.com/kdh6126/dkd-cross-cohort-audit (https://doi.org/10.5281/zenodo.23051697).

Use of AI tools. Claude Code (Anthropic) and ChatGPT (OpenAI) assisted with code development and refactoring, proposals for control analyses, language editing and revision of author-written text, and the code that draws the figures; no figure or result was generated by an AI model directly. Numerical results were computed from the cited datasets or explicitly described simulations using the analysis code. Automated checks were used to assess numerical consistency and provenance. The authors reviewed the code, analyses and manuscript and take responsibility for the final content.

The same statement appears in the manuscript.

The manuscript is original, is not under consideration elsewhere, and all authors have approved the submission. The authors declare no competing interests.

Thank you for considering our manuscript.

Yours sincerely

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Manuscript Information

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